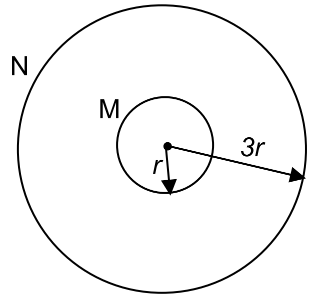
24	A flat circular coil M is mounted co-axially inside another flat circular coil N. N has three times the radius of M.						



The two coils are connected in series to the same power supply. At the centre of the two coils, the magnetic field created by M is opposite in direction to the magnetic field created by N.

What is the ratio of the number of turns of coil M to the number of turns of coil N to create zero magnetic flux density at the centre?

A	1 : 3	**B**	2 : 3	**C**	3 : 1	**D**	3 : 2

Answer: A

For flat circular coil, at its CENTRE: $B = \frac{\mu_o N I}{2R}$

For resultant flux density at the centre to zero for both coils,
B_N and B_M at the centre must be equal in magnitude.
In series implies same magnitude of current I flows in both coils.

$$B_N = B_M$$

$$\frac{\mu_o N_N I}{2(3r)} = \frac{\mu_o N_M I}{2(r)}$$

$$\frac{N_N}{3} = \frac{N_M}{1}$$

$$\frac{N_M}{N_N} = \frac{1}{3}$$

25	In a velocity selection, positively charged particles $+q$ enter an electric field of strength E and